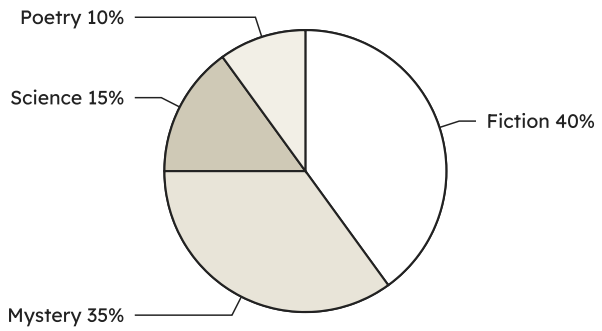
Benchmark: **MA.7.DP.1.4** · Circle graphs Student code: _____ Calculator: yes

Learning goal: I can create and interpret circle graphs. (circle your level after scoring)

4	I can construct a circle graph from data and use it to answer a comparison question.
3	I can find the sector angle or percent for each category and interpret a circle graph.
2	I can read a percent or category from a circle graph.
1	I need help to start this benchmark.

1. The circle graph shows how 540 people answered a survey.

How many more people chose Fiction than Science?



2. A survey asked 320 people which pet they would most like to have. The results are shown in the table.

Dogs	Cats	Fish	Birds
128	112	32	48

In a circle graph of the data, what percent of the circle is the Fish section?

3. 420 people were asked which color they like best. In a circle graph of the results, the Purple section has a central angle of 54°.

How many people chose Purple?

A. 15

B. 357

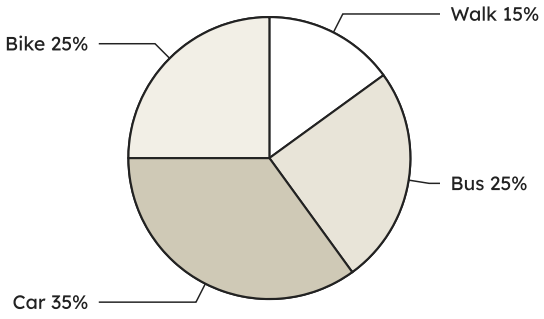
C. 63

D. 54

Turn over →

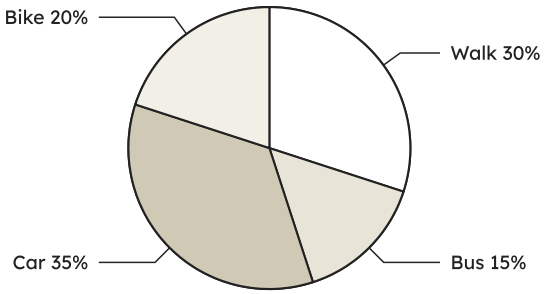
4. The circle graph shows how 740 students answered a survey.

How many students chose Bus?



5. 360 students were asked how they get to school. The circle graph shows the results.

Select **all** the statements that the circle graph supports.



Select **all** the correct answers.

A. ☐ Walk was chosen by twice as many students as Car.

B. ☐ Bus and Bike together were chosen by more than half of the students.

C. ☐ 18 more students chose Car than Walk.

D. ☐ The Bike section has a central angle of 20°.

E. ☐ 54 of the 360 students chose Bus.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-DP.1.4 · MA.7.DP.1.4 · Circle graphs

Unit 7: Percentage Problems With Proportional Relationships. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **135 more people**

Fiction: $0.4 \times 540 = 216$ people.

Science: $0.15 \times 540 = 81$ people.

$216 - 81 = 135$ more people chose Fiction.

2. Regular **10%**

Fish was chosen by 32 of the 320 people: $\frac{32}{320} = \frac{1}{10} = 10\%$.

The Fish section is 10% of the circle.

3. Regular **C. 63**

The section is 54° of the whole circle, 360°: $\frac{54}{360} = \frac{3}{20} = 15\%$.

So 15% of the 420 people chose Purple: $0.15 \times 420 = 63$ people.

4. Regular **185 students**

Bus is 25% of the circle.

$0.25 \times 740 = 185$ students.

5. Rigor **C. 18 more students chose Car than Walk; E. 54 of the 360 students chose Bus.**

Counts: Walk $0.3 \times 360 = 108$; Bus $0.15 \times 360 = 54$; Car $0.35 \times 360 = 126$; Bike $0.2 \times 360 = 72$.

Central angles: Walk $30\% \text{ of } 360^\circ = 108^\circ$; Bus $15\% \text{ of } 360^\circ = 54^\circ$; Car $35\% \text{ of } 360^\circ = 126^\circ$; Bike $20\% \text{ of } 360^\circ = 72^\circ$.

True: 54 of the 360 students chose Bus. 18 more students chose Car than Walk.

Scoring: 1 point only if every correct box is checked and no other box is; anything else earns 0.

Re-check strip (3 items, C-DP.1.4-R)

1. Regular **36°**

Bus was chosen by 22 of the 220 students: $\frac{22}{220} = \frac{1}{10} = 10\%$.

A whole circle is 360°, so the central angle is $10\% \text{ of } 360^\circ = 0.1 \times 360 = 36^\circ$.

2. Regular **D. 60**

The section is 90° of the whole circle, 360°: $\frac{90}{360} = \frac{1}{4} = 25\%$.

So 25% of the 240 students chose Pizza: $0.25 \times 240 = 60$ students.

3. Regular **23 more people**

Summer: $0.2 \times 460 = 92$ people.

Spring: $0.15 \times 460 = 69$ people.

$92 - 69 = 23$ more people chose Summer.

Comparison strip (2 items, C-DP.1.4-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **15%**

$\frac{3}{20} = 15\%$ and $\frac{2}{5} = 40\%$.

A circle graph always totals 100%: $15 + 30 + 40 = 85$.

$100 - 85 = 15\%$.

2. Regular **C. 16**

The section is 36° of the whole circle, 360°: $\frac{36}{360} = \frac{1}{10} = 10\%$.

So 10% of the 160 students chose Music: $0.1 \times 160 = 16$ students.

Reteach plan — C-DP.1.4

MA.7.DP.1.4 · Circle graphs

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

1. A survey asked 220 students how they get to school. The results are shown in the table.

Walk	Bicycle	Car	Bus
44	110	44	22

In a circle graph of the data, what is the measure of the central angle of the Bus section?

2. 240 students were asked which lunch they like best. In a circle graph of the results, the Pizza section has a central angle of 90°.

How many students chose Pizza?

A. 25

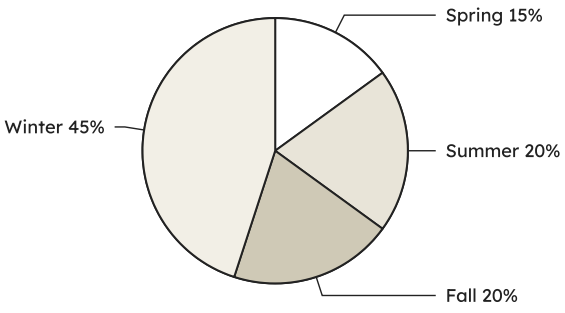
B. 90

C. 180

D. 60

3. The circle graph shows how 460 people answered a survey.

How many more people chose Summer than Spring?



Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-DP.1.4-X

Benchmark: **MA.7.DP.1.4** · Circle graphs Student code: _____ Calculator: yes

1. A circle graph of survey results has four sections: Cats, Dogs, Birds and Fish. The Cats section is $\frac{3}{20}$ of the graph, the Dogs section is 30%, and the Birds section is $\frac{2}{5}$ of the graph. The percent for Fish is not given.

What percent of the circle graph is Fish?

2. 160 students were asked what they like to do after school. In a circle graph of the results, the Music section has a central angle of 36°.

How many students chose Music?

A. 36 **B.** 144

C. 16 **D.** 10

Score: ____ / 2 → copy to your progress strip

7th Grade Math · Comparison items (2) C-DP.1.4-X

Benchmark: **MA.7.DP.1.4** · Circle graphs Student code: _____ Calculator: yes

1. A circle graph of survey results has four sections: Cats, Dogs, Birds and Fish. The Cats section is $\frac{3}{20}$ of the graph, the Dogs section is 30%, and the Birds section is $\frac{2}{5}$ of the graph. The percent for Fish is not given.

What percent of the circle graph is Fish?

2. 160 students were asked what they like to do after school. In a circle graph of the results, the Music section has a central angle of 36°.

How many students chose Music?

A. 36 **B.** 144

C. 16 **D.** 10

Score: ____ / 2 → copy to your progress strip